

# AMARDEEP SAHA

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LeetCode | GitHub | LinkedIn | Portfolio

## EDUCATION

**Indian Institute of Information Technology , Ranchi**

Computer Science and Engineering B.Tech.

CGPA: 8.45

Ranchi, India

July 2019 - May 2023

## EXPERIENCE

**Indiamart | Backend Developer**

Noida, India | June 2023 - Present

- Designed and implemented a WhatsApp-based enquiry generation pipeline that identified user intent for specific modules and triggered automated engagement, resulting in INR 1 lakh additional profit, scaled onboarding by 200+ additional users per day using Golang, Kafka, WhatsApp Business API, PostgreSQL, Cassandra, CDC, and Cron Jobs.
- Architected a high-throughput concurrent caching layer in Golang with in-memory storage and Redis, cutting database calls by 90%+ and achieving 1ms p95 latency with near-zero steady-state reads.
- Built Kafka-driven backend pipelines for location validation and enrichment with PostgreSQL and CDC, detecting anomalies beyond 250 km and triggering asynchronous remediation workflows.
- Led backend performance optimization across 5+ service SRS reviews, replacing sequential workflows with async processing to reduce page load time by 25%.
- Developed SSR and search-indexability improvements with React and Node.js, increasing organic traffic by 5% through better SEO and metadata handling.
- Owned mobile enquiry enrichment workflows with intelligent extraction, raising valid form fill rates by 10% and improving downstream conversion.
- Implemented Server-Side Rendering (SSR) using React and Node.js to enhance SEO, improve search engine indexability, and drive a 5% increase in organic traffic.
- Engineered scalable, data-intensive frontend architectures capable of handling 200+ concurrent asynchronous requests per component while maintaining UI responsiveness, stability, and performance under high load.
- Owned and optimized automated data enrichment workflows for mobile user enquiries, increasing valid form fill rates by 10% through intelligent extraction and enrichment strategies.
- Integrated a sentiment analysis capability using HuggingFaces multilingual BERT model to evaluate user tone and intent, enabling smarter downstream processing and enhanced user interaction analysis.

**CoantumLeap | Research and Development Intern**

Remote, India | March 2022 - August 2022

- Engineered a full-stack DAO application to manage over 10,000 members and transactions.
- Optimized gas-efficient smart contracts in Solidity for token sharing and stake distribution logic.
- Implemented frontend-to-blockchain connectivity using Ethers.js and MetaMask.
- Deployed smart contracts to the Ethereum blockchain with Remix IDE.

## SKILLS

|                        |                                                                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Programming Languages: | JavaScript, HTML, PHP, C, C++, Bash, TypeScript, Solidity, Python, Golang, SQL                                                        |
| Libraries/Frameworks:  | React, Node, Express, Bootstrap, Tailwind, Webpack, Next, Yii2, REST API, Redux, Ethers, Pandas, Babel, Jest, Next, Apache Kafka, Gin |
| Tools / Platforms:     | Git, VS Code, Nano, Remix, Ethereum, Docker, Devtron                                                                                  |
| Databases:             | MongoDB, SQL, PostgreSQL, Redis                                                                                                       |

## PROJECTS / OPEN-SOURCE

**Video Conferencing App | Link**

React JS, WebRTC , SocketIO, SQL , NodeJS, Redis, Docker

- Engineered a full-stack, real-time video conferencing application .
- Implemented peer-to-peer audio/video streaming with WebRTC and established real-time signaling and chat functionality using Socket.IO.
- Developed a secure backend featuring JWT-based authentication and a robust room management system, paired with a modern, responsive UI built with Tailwind CSS.

### **Voice-Enabled Task Tracker | Link**

*Next.js, TypeScript, Node.js, Express, SQLite, LLMs, Docker*

- Built a backend-driven task management system with REST APIs supporting CRUD, filtering, and search, using Node.js, Express, and SQLite with clear service separation.
- Designed an LLM-powered natural language processing pipeline using Groq SDK, converting voice input into structured task data (title, priority, due date, status).
- Integrated speech-to-text ingestion with backend services and a Next.js frontend, enabling real-time voice-based task creation and Kanban workflows.

### **Research Paper " A Survey Paper on Developments in Blockchain"**

- Authored and published an accepted research paper, "A Survey Paper on Developments in Blockchain," at IIIT Ranchi.
- Conducted a comprehensive literature review, analyzing over 25 research papers to evaluate the evolution of blockchain consensus mechanisms.